Course Code: CSE2101	**Course Name:** Data Analytics Using Python	
Credits: 3 (2-0-2)	**Contact Hours:** 2 hours Theory & 2 hours lab per week	
Batch: 2024, 3rd Sem **Academic Year:** 2025-26	**Semester Duration:** 29th July to 28th Nov 2025	
Course Faculty: **1. Dr. Kiran Sharma** Kiran.sharma@bmu.edu.in Cabin #87, 4th floor, E2 building Sections: CSE-1, CSE-2, CSE-3 & CSE-4 **2. Prof. Anantha Rao** anantha.rao@bmu.edu.in Cabin #35, 4th floor, E2 building Sections: CSE-7, CSE-8 **3. Ms. Anusha Chhabra** anusha.chhabra@ bmu.edu.in Cabin #, 4th floor, E2 building Sections: **4. Mr. Sudesh Bainsla** v-sudesh.bainsla@bmu.edu.in Cabin #, 4th floor, E2 building Sections:	**Course Coordinator:** Dr. Kiran Sharma **Email:** Kiran.sharma@bmu.edu.in **Office:** Cabin #87, 4th floor, E2 building	

Aim of the course:

The course aims to develop a solid understanding of data analytics by combining foundational programming skills with practical techniques for analyzing and interpreting data. Students will learn how to work with various types of data, prepare and clean datasets, and apply statistical thinking to uncover insights. Emphasis is placed on visualizing patterns, exploring relationships, and understanding variability in data. The course also introduces approaches for building and evaluating predictive models, along with methods to improve model performance and reduce complexity. By the end, students will be able to apply analytical thinking and computational tools to solve real-world problems using data.

Course Overview and Context:

This course offers a hands-on introduction to Python for data science, using Google Colab and Jupyter Notebook. It covers core concepts such as data structures, data manipulation with Pandas and NumPy, and data visualization using Matplotlib and Seaborn. Learners will explore types and levels of data, the data science lifecycle, data cleaning, transformation, and exploratory analysis. The course also introduces basic statistical concepts, hypothesis testing, and machine learning techniques including KNN, K-Means, PCA, and regression models, equipping students with foundational skills for real-world data analysis and modeling.

Prerequisite: Python Programming and Statistics

Course Outcomes (CO): At the end of the course the students should be able to do the following:

	Course Outcome Statement	Mapped Component	Associated Activities /Assessment
CO1	Understand basic concepts and visualize data to gain insight.	Concepts Knowledge	Planned quizzes/ viva and problem solving to understand knowledge gain
CO2	Apply data science skills to extract and interpret datasets.	Skill development	Hands-on Lab session, case studies
CO3	Analyse and interpret data using statistical models.	Employability	Assessment of concepts of real-life datasets and decision making

Topics of the course:

Topics	Number of Sessions (48)	Course Outcomes
Introduction to Python for Data Analytics: Setting up the environment: Google Colab and Jupyter Notebook; Basic syntax and programming constructs; Introduction to Python data structures (lists, tuples, dictionaries, sets)	6 (4 Th + 2 Lab)	CO1 & CO2
Data Handling and Manipulation: Introduction to Pandas; Reading and writing files; Loading and inspecting datasets; Data selection, filtering, and sorting; Working with NumPy for numerical operations	9 (6 Th + 3 Lab)	CO1, CO2 & CO3
Data Visualization: Introduction to data visualization principles; Using Matplotlib: Line plot, scatter plot, histogram, bar chart, area plot, box plot; Advanced visualizations with Seaborn: Heatmaps, pair plots, faceting, and trend analysis	6 (4 Th + 2 Lab)	CO1, CO2 & CO3
Fundamentals of Data Science: What is data science? Types and levels of data: Structured vs. unstructured data; Quantitative vs. qualitative data; Overview of the data science life cycle	1Th	CO1
Data Collection and Preparation: Techniques for data acquisition; Handling missing data and outliers; Data scrubbing and cleaning; Data transformation and normalization	3 (2 Th + 1 Lab)	CO2 & CO3
Exploratory Data Analysis (EDA): Summarizing and visualizing data distributions; Identifying trends and patterns; Comparing population vs. sample; Moments and generating functions; Measures of central tendency and variability	5 (3Th + 2 Lab)	CO2 & CO3
Statistical Foundations: Hypothesis formulation and testing; Understanding bias and variance; Significance levels and p-values; Confidence intervals	3 (2 Th + 1 Lab)	CO1& CO2
Machine Learning Basics: Introduction to supervised learning: Classification using K-Nearest Neighbors (KNN); Introduction to unsupervised learning: Clustering with K-Means; Defining and training models; Evaluating model performance	9 (6 Th + 3 Lab)	CO1, CO2 & CO3
Feature Engineering and Dimensionality Reduction: Techniques for feature extraction and selection; Introduction to Principal Component Analysis (PCA)	3 (2 Th + 1 Lab)	CO1, CO2 & CO3

Predictive Modeling with Regression: Concept of regression analysis; Building linear regression models; Logistic regression for classification tasks	3 (2 Th + 1 Lab)	CO1, CO2 & CO3

CO/PO Mapping: 1—Weak, 2—Moderate and 3-- High

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	1	3									2			
CO2	3	2	1	3									3	1		
CO3	3	2	1	3									3	1		

Learning Resources:

Textbook:

1) Sinan Ozdemir. Principles of Data Science: Learn the techniques and math you need to start making sense of your data. Packt Publishing.
2) VanderPlas, Jake. Python data science handbook: Essential tools for working with data. & OReilly Media, Inc.2023.

Reference Books:

1) Peng, Roger D., and Elizabeth Matsui. The Art of Data Science. A Guide for Anyone Who Works with Data Skybrude Consulting, 2015.
2) Foreman, John W. Data smart: Using data science to transform information into insight. John Wiley & Sons, 2014.
3) Haider, Murtaza. Getting Started with Data Science: Making Sense of Data with Analytics. IBM Press, 2015.

Note: Instructors will regularly post the necessary learning resources such as lecture resources to the online course management portal i.e., Maitri/Google-classroom.

Experiential Learning Component:

50% of course delivery will be based on hands-on exercises.

Assessment Pattern:

Component	Duration	Weightage (%)	Evaluation Week	Remarks	Mapped COs
Lab-I Evaluation	2hrs	30%	Sep 2nd to 3rd week in lab	Code-based implementation (20 marks) & Viva/Quiz (10 marks)	CO1 & CO2
Lab-II Evaluation	2 hrs	30%	Nov 2nd to 3rd Week in lab	Code-based implementation (20 marks) & Viva/Quiz (10 marks)	CO1, CO2 & CO3
End-Sem Exam	2hrs	40%	As per university	Close book written exam	CO1, CO2 & CO3

			calendar		

Student Responsibilities:

- Attend lectures and do the work Lab Assignments as per instructions.
- Participate in the discussions/assignments held during classes.
- Check announcements on Google Classroom and emails regularly.
- Submit the assigned task on time.
- Regularly check marks on Maitri to ensure they are up to date.
- Participate in class and take necessary actions to grasp the material. Asking questions is encouraged.
- Communicate any concerns by speaking directly with the instructor.

Attendance Policy: Students are expected to attend classes regularly. Failure to follow the classes regularly and adhere to the expected attendance percentage will result in losing quiz/lab marks and a reduction of the grade as per the University's grading policy.

Recourse Examination Policy: In case a student fails the course, a one-time recourse is permitted as per the academic regulations of the University. Recourse is allowed only for the End Semester examination.

Make-up policy: No make-up exam will be conducted for unexcused absences. The faculty needs to be informed in advance in case the student is not going to appear for any evaluation component, and it is at the discretion of the faculty to sanction makeup for an evaluation component.

Behavior Expectations: No mobile phones and other distractive gadgets are permitted in the class.

Academic Dishonesty/Cheating/Plagiarism: Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.